CS roadmap

Computer Science & Engineering (source: CourseDetails_CS.pdf, 296 entries).

How to use a course file

1. Watch the 🎓 **Main course** playlist in order, following the **Study plan** table (it links each week to the exact lecture videos).
2. If a lecturer doesn't work for you, switch to one of the 🎥 **Other full lecture series**; use ⚡ **Short videos** for intuition before or after a hard topic.
3. Read the book chapters and do the **Practice** project. Watching alone is not learning.

About the links: every YouTube playlist was opened on 2026-10-22; its title, video count and lecture list are copied from YouTube, not typed by hand. Official channels (NPTEL, IIT Madras, MIT OCW, Stanford Online, CMU, Harvard and others) were preferred; uploads by other people are marked *re-upload*. In the Books section, ✅ means the link was checked in a web search; other book/website links were not re-checked, so confirm they are still online.

Tier A = core, do it. Tier B = recommended elective. Tier C = advanced, only after its prerequisites.

Order of courses

Phase 1

Course	Tier	Needs first	Effort	Main video course	File
CS1111 Problem Solving using Computers	A	Nothing	6-8 weeks, ~6 h/week	Introduction to C Programming	courses/P1_CS1111_problem-solving-using-computers.pdf
CS1200 Discrete Mathematics for Computer Science	A	School maths only	10-12 weeks, ~6 h/week	Computer Sc - Discrete Mathematical Structures	courses/P1_CS1200_discrete-mathematics-for-computer-science.pdf
CS1234 Small-Scale Application Development (Linux + tooling)	A	CS1111 (basic C)	4-6 weeks, ~5 h/week	Missing Semester IAP 2020	courses/P1_CS1234_small-scale-application-development-linux-toolin.pdf

Phase 2

Course	Tier	Needs first	Effort	Main video course	File
CS2705 Programming and Data Structures	A	CS1111	10-12 weeks, ~8 h/week	Computer Sc - Programming and Data Structure	courses/P2_CS2705_programming-and-data-structures.pdf
CS1202 Discrete Mathematics II (graphs, algebra, number theory)	A	CS1200	8-10 weeks, ~5 h/week	MIT 6.042J Mathematics for Computer Science, Fall 2010	courses/P2_CS1202_discrete-mathematics-ii-graphs-algebra-number-th.pdf
CS2222 Advanced Programming Techniques (deep C)	B	CS1111, CS1234	6-8 weeks	CS50x 2025 Lectures	courses/P2_CS2222_advanced-programming-techniques-deep-c.pdf

Phase 3

Course	Tier	Needs first	Effort	Main video course	File
CS2300+CS2310 Foundations of Computer Systems Design (+ lab)	A	CS1111, CS1200	12 weeks, ~8 h/week	Foundations to Computer Systems Design	courses/P3_CS2300_CS2310_foundations-of-computer-systems-design-lab.pdf
CS2800+CS2810 Design and Analysis of Algorithms (+ OO lab)	A	CS2705, CS1200, CS1202	14 weeks, ~9 h/week	Design and analysis of algorithms	courses/P3_CS2800_CS2810_design-and-analysis-of-algorithms-oo-lab.pdf
CS2600+CS2610 Computer Organization and Architecture (+ lab)	A	CS2300 (or at least gates + assembly idea)	12 weeks	Computer Sc - Computer Organization	courses/P3_CS2600_CS2610_computer-organization-and-architecture-lab.pdf
CS2200 Language, Machines and Computations (theory of computation)	A	CS1200	12 weeks, ~7 h/week	Computer - Theory of Automata, Formal Languages and Computation	courses/P3_CS2200_language-machines-and-computations-theory-of-com.pdf

Course	Tier	Needs first	Effort	Main video course	File
CS6015 Linear Algebra and Random Processes	A	School maths	12 weeks	MIT 18.06 Linear Algebra, Spring 2005	courses/P3_CS6015_linear-algebra-and-random-processes.pdf

Phase 4

Course	Tier	Needs first	Effort	Main video course	File
CS2320 Introduction to Graph Theory	B	CS1202	10 weeks	Computer - Graph Theory	courses/P4_CS2320_introduction-to-graph-theory.pdf
CS3500+CS3510 Operating Systems (+ lab)	A	CS2705, CS2600, CS2222 (C)	14 weeks, ~9 h/week	Introduction to Operating Systems / IIT Madras	courses/P4_CS3500_CS3510_operating-systems-lab.pdf
CS3300+CS3310 Compiler Design (+ lab)	A	CS2200, CS2705, CS2600	14 weeks, ~9 h/week	NOC Jan 2019: Compiler Design	courses/P4_CS3300_CS3310_compiler-design-lab.pdf
CS3700 Introduction to Database Systems	A	CS2705, CS1200	10-12 weeks	Introduction to Database Systems	courses/P4_CS3700_introduction-to-database-systems.pdf
CS3205+CS3210 Introduction to Computer Networks (+ lab)	A	CS2705, CS3500 helps	12 weeks	Computer Networks and Internet Protocol	courses/P4_CS3205_CS3210_introduction-to-computer-networks-lab.pdf

Phase 5

Course	Tier	Needs first	Effort	Main video course	File
CS3400+CS3410 Principles of Software Engineering (+ lab)	B	CS2705, CS1234	8-10 weeks	Software Engineering by Prof. Rajib Mall	courses/P5_CS3400_CS3410_principles-of-software-engineering-lab.pdf
CS4100+CS4110 Computer System Design (+ lab)	B	CS2600, CS3500	10-12 weeks	Livestream - Computer Architecture - ETH Zürich (Fall 2023)	courses/P5_CS4100_CS4110_computer-system-design-lab.pdf
CS4011 Principles of Machine Learning	A	CS6015 or equivalent, CS2800	12-14 weeks, ~9 h/week	NPTEL MOOC Machine Learning 2016	courses/P5_CS4011_principles-of-machine-learning.pdf
CS6380 Artificial Intelligence	A	CS2800, CS6015	10-12 weeks	Artificial Intelligence Search Methods For Problem Solving	courses/P5_CS6380_artificial-intelligence.pdf

Phase 6

Course	Tier	Needs first	Effort	Main video course	File
CS6910 Fundamentals of Deep Learning	A	CS4011, calculus	12 weeks, ~10 h/week	Deep Learning	courses/P6_CS6910_fundamentals-of-deep-learning.pdf
CS6848 Principles of Programming Languages	B	CS2200, CS3300	10-12 weeks	Computer Sc - Principles of Programming Languages	courses/P6_CS6848_principles-of-programming-languages.pdf
CS6550 Database Management System (advanced, with CS3700)	B	CS3700	10-12 weeks	CMU Advanced Database Systems (15-721 / Spring 2024)	courses/P6_CS6550_database-management-system-advanced-with-cs3700.pdf
CS6500 Network Security / security cluster	B	CS3205, CS1202	10-12 weeks	Computer - Cryptography and Network Security	courses/P6_CS6500_network-security-security-cluster.pdf
CS6720 Data Mining	B	CS3700, CS4011	10 weeks	Data Mining for Decision Making	courses/P6_CS6720_data-mining.pdf

Phase 7

Course	Tier	Needs first	Effort	Main video course	File
CS6700 Reinforcement Learning	B	CS4011, probability	12 weeks	Reinforcement Learning	courses/P7_CS6700_reinforcement-learning.pdf

Course	Tier	Needs first	Effort	Main video course	File
CS6370 Natural Language Processing	B	CS4011, CS6910	12 weeks	Stanford CS224N Natural Language Processing with Deep Learning I Spring 2024 I Professor Christopher Manning	courses/P7_CS6370_natural-language-processing.pdf
CS6350 Computer Vision	B	CS6910, linear algebra	12 weeks	Stanford CS231N Deep Learning for Computer Vision I 2025	courses/P7_CS6350_computer-vision.pdf
CS6847 Cloud Computing + Distributed Computing	B	CS3500, CS3205, CS3700	12 weeks	MIT 6.824 Distributed Systems (Spring 2020)	courses/P7_CS6847_cloud-computing-distributed-computing.pdf
CS6741 Statistical Foundations of Data Science + related stats/optimisation	C	CS6015, CS4011	12 weeks	Statistics 110: Probability	courses/P7_CS6741_statistical-foundations-of-data-science-related-.pdf
CS6596 Parallel and concurrent programming (CS6596, CS6868)	C	CS3500, CS2600	10 weeks	Stanford CS149 I Parallel Computing I 2023 I Kayvon Fatahalian and Kunle Olukotun	courses/P7_CS6596_parallel-and-concurrent-programming-cs6596-cs686.pdf

Phase 8

Course	Tier	Needs first	Effort	Main video course	File
CS6030 Logic and Combinatorics for CS	C	CS1200, CS1202	10-12 weeks	Mathematics - Mathematical Logic	courses/P8_CS6030_logic-and-combinatorics-for-cs.pdf
CS6886 Systems Engineering for Deep Learning	C	CS6910, CS2600, CS3500	10-12 weeks	EfficientML.ai Course / 2023 Fall / MIT 6.5940	courses/P8_CS6886_systems-engineering-for-deep-learning.pdf

Not doable (or not worth doing) as self-study

Course(s)	Why
INSA, *INS* (Analog devices, Semiconductor devices ...)	Interdisciplinary courses with empty syllabus text in the PDF; nothing to plan from.
RC, *RCCS*, Research credits / Comprehensive exam / Seminar 2	Not courses; they record research work.
CS4900, CS4910, CS5705, CS5715, CS5980/5990/6000, CS5931/5932, CS5998, CS6008/6019/6119, ID5490-93, CS6900	Projects and thesis credits, assigned by a supervisor.
CS3660, CS3666 Industrial Training	Needs an internship.
CS4999 Seminar, CS7999, CS6190, CS7xxx Topics/Recent topics courses	Rotating research-paper seminars; no fixed syllabus.
CS6853 GIAN, CS8852	One-off visiting-faculty courses.
NUS* courses	Exchange-programme equivalents of the same subjects; use the main CS course instead.
CS6840, CS6845, CS6850, CS6240, CS6122, CS6101, CS6170, CS6841, CS6025, CS6535, CS6525, CS6780	Research-level theory (complexity, pseudorandomness, approximation, parameterised algorithms). Do CS2800 + CS2200 + CS1202 well first; then pick one.

Course(s)	Why
CS6760W, CS6230W, CS6330, CS6620W, CS6640	VLSI/testing/advanced architecture; only if you plan hardware work.
Remaining PG electives (wireless, IoT, quantum, blockchain, ontologies ...)	Doable later if you want that niche; no roadmap written now.